

EMS Optimization — Phase Comparison Summary

TL;DR: 12 phases originally planned, 7 phases implemented

The original project outline defined **12 sequential sections/phases**. The implementation consolidated these into **7 execution phases** that fully cover all 12 original sections (verified at 100% alignment).

Original Phases Planned (from EMS_Project_Outline_Hybrid_Final_Submission.docx)

The outline document defines 12 numbered sections, each with explicit “Main tasks” and a “Completion output”:

#	Original Phase	Description
1	Project Purpose & Decision Problem	Define the operational decision, confirm baseline vs. alternative policies, frame as decision-support study
2	Research Questions & Measures of Effectiveness	Finalize research questions, choose primary/secondary MOEs, define service-level thresholds, specify decision rules
3	Study Scope & System Boundary	Lock geography (Manhattan + CBD), define spatial units, list included/excluded model components
4	Data Strategy & Engineering Plan	Acquire crash records, build zone-by-time panel, create site-to-zone distance matrix, document data pipeline
5	GIS & Spatial Analysis Plan	Select spatial units, produce hotspot/density maps, validate firehouse-to-zone matrices
6	Input Modeling & Crash-Demand Estimation	EDA of crash counts, estimate arrival process (NHPP), export simulation-ready lambda tables
7	Allocation Policy Design & Scenario Library	Define baseline + alternative policies, formulate MIP allocation, generate scenario library
8	Conceptual Simulation Model	Define entities, resources, queues, events, state variables, logic flow; create conceptual diagram
9	Simulation Implementation Plan	Build SimPy DES engine (data loaders, dispatch, queue handling, output logging)
10		Code verification (toy examples, unit tests), validation

#	Original Phase	Description
	Verification & Validation Plan	(baseline comparison, sensitivity checks)
11	Experimental Design & Output Analysis	Factorial design, production runs, ANOVA, confidence intervals, CBD robustness comparison
12	Recommendation, Documentation & Final Reporting	Final report, presentation, reproducible code package, policy recommendation

Phases Implemented (from `final_summary.md` and `project_archive.md`)

The project was executed in **7 phases**, each completed over approximately one week:

Phase	Description	Status	Key Deliverables	Original Sections Covered
Phase 1	Data Processing & EDA	✓ Complete	628K Manhattan crashes, 48 fire-houses, 10 EDA figures	Sections 1-5 (problem definition, scope, data, GIS)
Phase 2	Demand & Service Modeling	✓ Complete	NHPP model ($\lambda_0=3.48/\text{hr}$), lambda tables, distance matrix	Section 6 (input modeling)
Phase 3	Optimization Models	✓ Complete	3 MIP formulations (Demand-Weighted, P-Median, Maximal Coverage), policy library	Section 7 (allocation policy design)
Phase 4	DES Simulation + V&V	✓ Complete	SimPy engine (7 modules), 39 unit tests, verification log	Sections 8-10 (conceptual model, implementation, V&V)
Phase 5	Production Experiments	✓ Complete	1,770 runs across 5 experiments (incl. CBD robustness)	Section 11 (experimental design)
Phase 6	Statistical Analysis	✓ Complete	ANOVA, Tukey HSD, Cohen's d, 5 pub-quality figures, 4 tables	Section 11 (output analysis)
Phase 7	Final Report & Docs	✓ Complete	Technical report, executive presentation, implementation roadmap, archive	Section 12 (reporting & documentation)

Mapping: Original 12 Sections → Implemented 7 Phases

Original Outline	Implementation
1. Project Purpose & Decision Problem	Phase 1: Data Processing & EDA
2. Research Questions & MOEs	
3. Study Scope & System Boundary	
4. Data Strategy & Engineering Plan	
5. GIS & Spatial Analysis Plan	
6. Input Modeling & Demand Estimation	Phase 2: Demand & Service Modeling
7. Allocation Policy Design	Phase 3: Optimization Models
8. Conceptual Simulation Model	Phase 4: DES Simulation + V&V
9. Simulation Implementation	
10. Verification & Validation	
11. Experimental Design & Output	Phase 5: Production Experiments
	Phase 6: Statistical Analysis
12. Recommendation & Reporting	Phase 7: Final Report & Docs

Discrepancies & Observations

Consolidation (not omission)

- **No original sections were dropped.** The `project_alignment_verification.md` confirms 100% coverage across all 12 outline sections.
- The 12 outline sections were **consolidated into 7 execution phases** for practical workflow efficiency.

Key consolidations:

1. **Sections 1-5 → Phase 1:** Problem framing, scope, data acquisition, and spatial analysis were executed together as a single data-processing sprint.
2. **Sections 8-10 → Phase 4:** Conceptual modeling, implementation, and V&V were treated as one integrated simulation-development phase.
3. **Section 11 → Phases 5 + 6:** The outline’s single “Experimental Design & Output Analysis” section was **split into two phases** — production runs (Phase 5) and statistical analysis (Phase 6) — giving more rigor to each.

Additions beyond the original outline:

- **CBD Robustness Experiment** (Exp 5): 330 additional runs specifically for CBD-focused analysis — the outline mentioned CBD as a robustness check but the implementation gave it a full dedicated experiment.
- **Queue Analysis** (`queue_analysis.md`): Dedicated queueing-theory analysis not explicitly called out as a standalone deliverable in the outline.
- **Implementation Roadmap** (`implementation_roadmap.md`): A 3-phase real-world deployment plan that goes beyond the academic scope of the original outline.
- **Gap Closure Report** and **Gap Remediation Plan**: Quality-assurance documents ensuring all outline requirements were met.

Final Answer

12 phases originally planned, 7 phases implemented — with all 12 original sections fully covered (100% alignment verified). The reduction from 12 to 7 reflects consolidation of related activities, not omission. One original section (Experimental Design & Output Analysis) was actually expanded into two separate phases for greater rigor.